

September 4, 2025

MIR-25-34

## Contact of Ferry *Ray Stoker, Jr.* with Floating Dock

On September 10, 2024, about 1823 local time, the vehicle and passenger ferry *Ray Stoker, Jr.* experienced a loss of propulsion control while underway in Galveston Channel, struck a floating dock at US Coast Guard Base Galveston, in Galveston, Texas, and grounded in about 4 feet of water (see figure 1 and figure 2).<sup>1</sup> Six crewmembers, seven passengers, and three vehicles were on board the ferry. There were no injuries, and no pollution was reported. Damage to the vessel and dock was estimated at \$373,800.<sup>2</sup>



**Figure 1.** *Ray Stoker, Jr.* docked in Galveston, Texas, on September 25, 2024, after striking the floating dock at the Coast Guard Base Galveston.

<sup>1</sup> In this report, all times are central daylight time, and all miles are statute miles.

<sup>2</sup> Visit [nts.gov](https://www.nts.gov) to find additional information in the [public docket](#) for this NTSB investigation (case no. DCA24FM062).

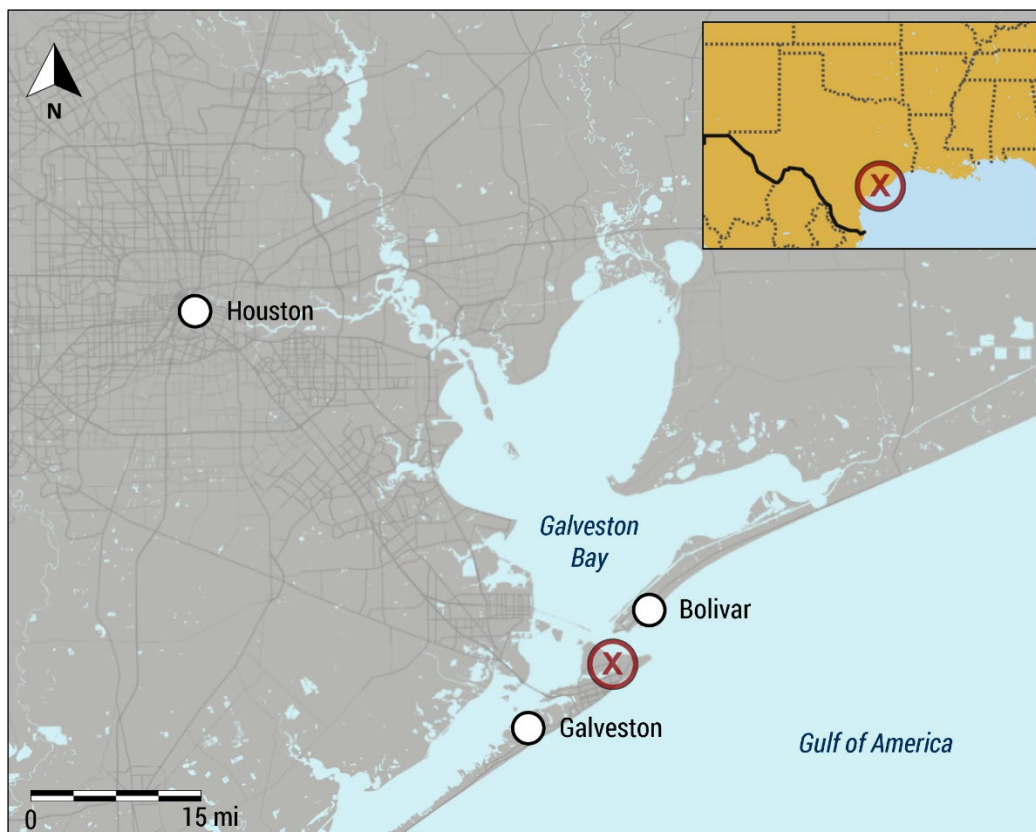
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**Casualty Summary**

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|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Casualty type</b>        | Contact                                                                                                               |
| <b>Location</b>             | Galveston Channel, US Coast Guard Base Galveston, Galveston, Texas<br>29°19.9' N, 094°46.36' W                        |
| <b>Date</b>                 | September 10, 2024                                                                                                    |
| <b>Time</b>                 | 1823 central daylight time (coordinated universal time -5 hrs)                                                        |
| <b>Persons on board</b>     | 13                                                                                                                    |
| <b>Injuries</b>             | None                                                                                                                  |
| <b>Property damage</b>      | \$373,800 est.                                                                                                        |
| <b>Environmental damage</b> | None                                                                                                                  |
| <b>Weather</b>              | Visibility 10 mi, overcast, winds from the northeast 10 kts, air temperature 80°F, sunset 1932, evening twilight 1956 |
| <b>Waterway information</b> | Channel; depth about 45 ft (center)                                                                                   |

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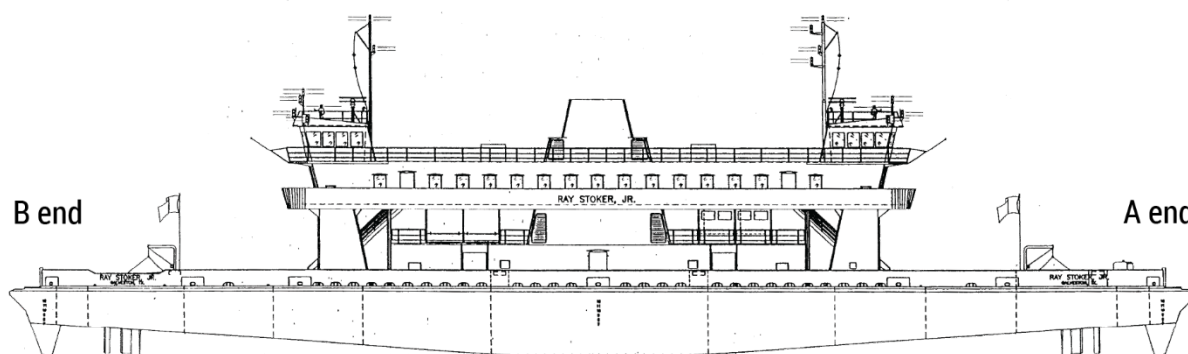
**Figure 2.** Area where the *Ray Stoker, Jr.* struck the floating dock at Coast Guard Base Galveston, as indicated by a circled X. (Background source: Google Maps)

# 1 Factual Information

## 1.1 Background

The 263-foot-long, welded-steel, open-deck vehicle and passenger ferry, *Ray Stoker, Jr.*, owned and operated by the Texas Department of Transportation, was built at Moss Point Marine shipyard in Mississippi in 1996. The ferry provided free, 24-hour vehicle and passenger service between Bolivar Peninsula, Texas, and Galveston Island, Texas, and had a maximum capacity of 70 vehicles.

The *Ray Stoker, Jr.* was a double-ended ferry with identical wheelhouses and propulsion units at each end (see figure 3). This configuration allowed the vessel to approach and leave its slips without turning around. Typically, the ferry's "A end" docked in Bolivar; the "B end" docked in Galveston. Watchstanding deck crewmembers repositioned themselves from one end to the other between transits. Each round trip began at the Galveston terminal, where the ferry was docked when not in use.



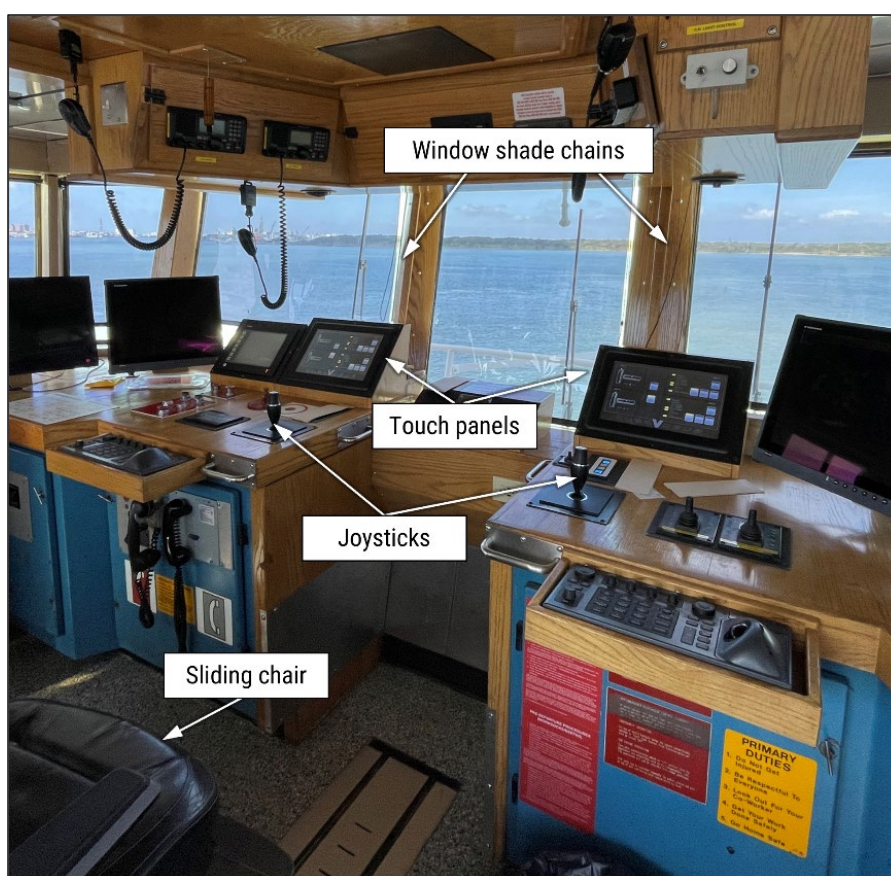
**Figure 3.** Profile view of *Ray Stoker, Jr.*, showing starboard side view of vessel with A end (Bolivar) and B end (Galveston) labeled. (Background source: Texas Department of Transportation)

Two cycloidal propellers, one at each end of the vessel, provided propulsion (thrust and steering control).<sup>3</sup> Each propeller was mechanically driven by a 12-cylinder marine diesel engine. The propulsion control system was upgraded

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<sup>3</sup> A *cycloidal propeller* consists of a large, rotating, horizontally configured circular disk beneath the vessel, driven by the main propulsion engine. The disk is fitted with a series of controllable pitch blades protruding vertically downward from the disk. The magnitude and direction of thrust are determined by the rotational speed of the disk, and the direction of thrust is determined by the positioning of the blade angles.

in 2020 from a mechanical to an electronic system. There were control stations in each of the wheelhouses and in the engine operating station (EOS), as well as a local emergency control station at each propulsion unit. During typical operations, the propulsion system was controlled from one of the wheelhouses. The wheelhouse control system consisted of two joysticks mounted on the operating console: one positioned by the operator's left hand, and one by the operator's right hand on either side of a sliding chair (see figure 4). The joysticks could be operated in synchronized or manual mode. When in synchronized mode, the right joystick controlled both the forward and aft propulsion units, and the left joystick controlled the side to side (athwart) movement of the forward unit. In manual mode, the left joystick controlled the forward propulsion unit, and the right joystick controlled the aft unit.



**Figure 4.** "B end" wheelhouse of *Ray Stoker, Jr.* ("A end" similar).

An angled touch panel was located forward of each joystick on the console, between the joysticks and the wheelhouse windows. The two touch panels (in each wheelhouse and in the EOS) provided system status, operational information, and system alarms. The ferry operator could make operational changes to the vessel's propulsion system on the touch panels, activating icons to change views, change operating parameters, adjust engine speed (idle, medium, high), or switch operating

mode from synchronized to manual; each panel also included a “station freeze” function (which would freeze the joystick setpoints of the active stations at their most recent settings). When the station freeze icon was activated, a solid, yellow indicating light illuminated above the icon.

Each window in the wheelhouse had an adjustable tinted shade mounted above it; each shade could be lowered or raised by a plastic chain that had a freely hanging plastic clip at the bottom.

## 1.2 Event Sequence

On September 10, at 1100, a crew of six, consisting of a captain (who had over 25 years in the maritime industry and about 5 months with the ferries), three deckhands, a chief engineer, and an oiler, reported to the vessel. The captain and deckhand went to the wheelhouse, and the chief engineer and oiler stood watch in the engine room.

At 1144, the *Ray Stoker, Jr.* departed Galveston for the first 2.7-mile round trip of the day between Galveston and Bolivar. Throughout the rest of the day, the crew completed 13 trips across Galveston Bay, averaging about 18 minutes for each round trip, transiting at 10 knots. The largest complement of the day was the second trip from Bolivar with one truck, 46 cars, and 144 passengers.

At 1810, the *Ray Stoker, Jr.* departed Bolivar for the fourteenth and final trip of the day, bound for Galveston with seven passengers and three cars aboard. The captain controlled the ferry from the “B end” wheelhouse, while the deckhand completed paperwork. The propulsion system was being operated in synchronized mode. The joysticks were used to input commands to control the pitch of the propulsion system, providing longitudinal and transverse thrust for propulsion and steering. The captain stated that, at this time, skies were overcast, with winds about 10-15 knots from the northeast, and there was no appreciable current.

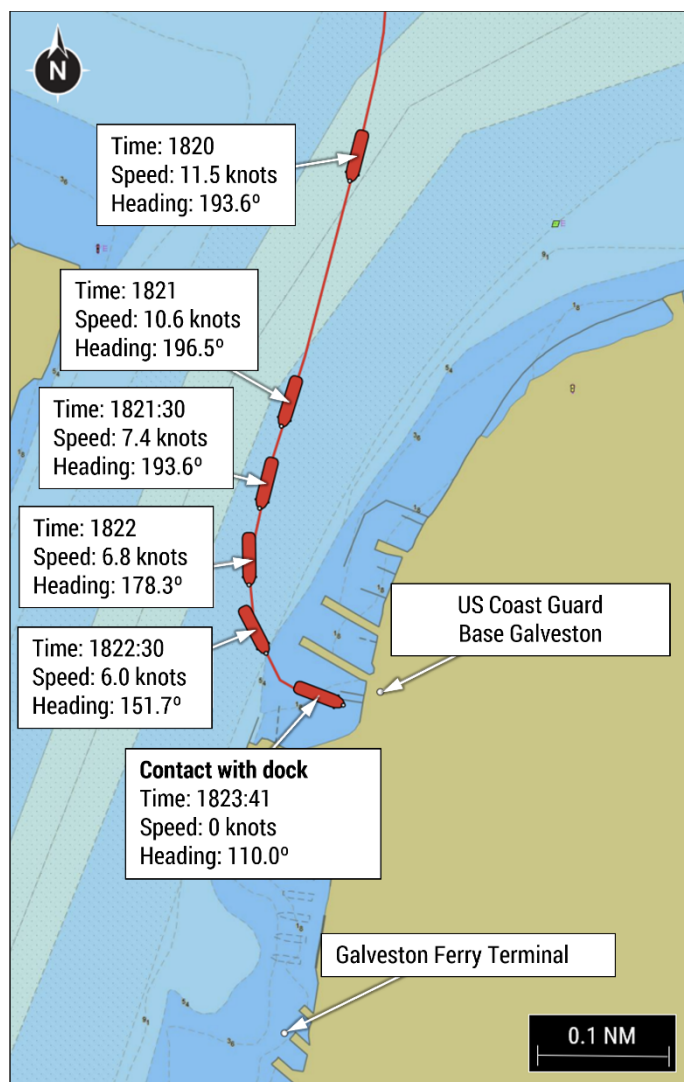
As the ferry approached its dock, at 10 knots in the Galveston Channel, with Coast Guard Base Galveston about 300 feet off its port side, the captain adjusted both joysticks for the propulsion system to their neutral positions to slow the vessel to about 8 knots and to maintain its current heading. Once the vessel’s speed was reduced to 8 knots, the captain adjusted the joysticks to re-apply power and maintain 8 knots at the same heading. However, the vessel began turning to port toward the Coast Guard base. The captain tried to correct the steering using both joysticks, but there was no response from either joystick. According to the captain, there were no alarms or abnormalities with the control system. The vessel slowed in speed but continued turning to port toward the Coast Guard docks.

The captain called the chief engineer in the EOS and advised him of the loss of control. At the chief engineer's instruction, the captain clicked the command icon to verify his station was in control and in synchronized mode, but there was no change in system status.

The chief engineer told investigators that he had experienced a similar loss of propulsion control on another vessel equipped with the same propulsion control system. He had rectified the situation by taking command control in the EOS and sending control back to the captain in the wheelhouse. He referred to this process as a "system reset." (He stated that the previous loss of control was caused by an overheated controller; the condition had been repaired on the *Ray Stoker, Jr.* and all similar ferries by installing air conditioning units in the controllers.)

Using the touch panel icons in the EOS, the chief engineer transferred control of the propulsion system down to the EOS, lowered the engines to idle speed, and transferred the control back to the wheelhouse (after determining the system was working properly from the EOS). The chief engineer told investigators that there had been no alarms or any indications of issues with the system. While the chief engineer was transferring the propulsion system, the captain used the public address system to warn passengers to brace for impact, and deck crewmembers notified passengers to brace themselves.

The deckhand left the wheelhouse and went down to the anchor and prepared to release it when directed. The ferry continued turning to port until it had turned about 90° and was parallel to a floating dock at the Coast Guard base. After the chief engineer transferred control of the propulsion system back to the wheelhouse, the captain regained control of the propulsion system, but about 1823, the *Ray Stoker, Jr.* contacted a floating dock on the west end of the Coast Guard base at 1 knot (see figure 5).



**Figure 5.** Trackline of *Ray Stoker, Jr.* from 1820 to 1823. (Background source: National Oceanic and Atmospheric Administration Electronic Navigation Chart US5TXTDB, as viewed on MadeSmart automatic identification system)

The ferry's portside rub rail pushed the first piling supporting the floating dock to the east, causing the piling to strike a metal railing and support structure of the adjacent fixed boat lift dock. The last piling for the floating dock was also pushed to the east and broke up the dock surrounding the piling (which was composed of concrete and foam). The ferry's lifeboat brackets rode up onto the floating dock, damaging the concrete and the metal structures holding it in place. The ferry then grounded in about 4 feet of water near the shoreline alongside the floating dock and came to a stop. The dock was not in use by any Coast Guard vessels or personnel at the time of the accident.

Immediately after the contact, crewmembers checked the *Ray Stoker, Jr.* for water ingress and determined there was none. The captain and chief engineer tested

propulsion control and verified the system was operating properly. The captain notified the Coast Guard and dispatch office and received permission to resume the transit to the ferry terminal.

The vessel tied up at 1900. There were no injuries reported by crewmembers or passengers. Drug and alcohol testing was conducted of all crewmembers; all results were negative.

## **1.3 Additional Information**

### **1.3.1 Damage**

After the casualty, divers surveyed the affected floating pier and noted broken concrete under the floating pier and damaged rollers attached to the pilings. (The floating pier had previously been damaged and was listing about 30–40° before being struck by the *Ray Stoker, Jr.*) The pier had an estimated replacement value of \$368,800. Weeks after the accident, a storm further damaged the floating pier, which led to its decommissioning and removal.

Damage to the *Ray Stoker, Jr.* consisted of scraped paint about 1 foot above the waterline along the port side of the hull. Additionally, the portside rub rail had about 3 feet of white paint transferred to it from the floating dock pilings. Steel brackets under the lifeboat launching area were scraped, dented, and broken. Damage was estimated at \$5,000. The brackets were repaired or replaced by the ferry's shoreside maintenance team.

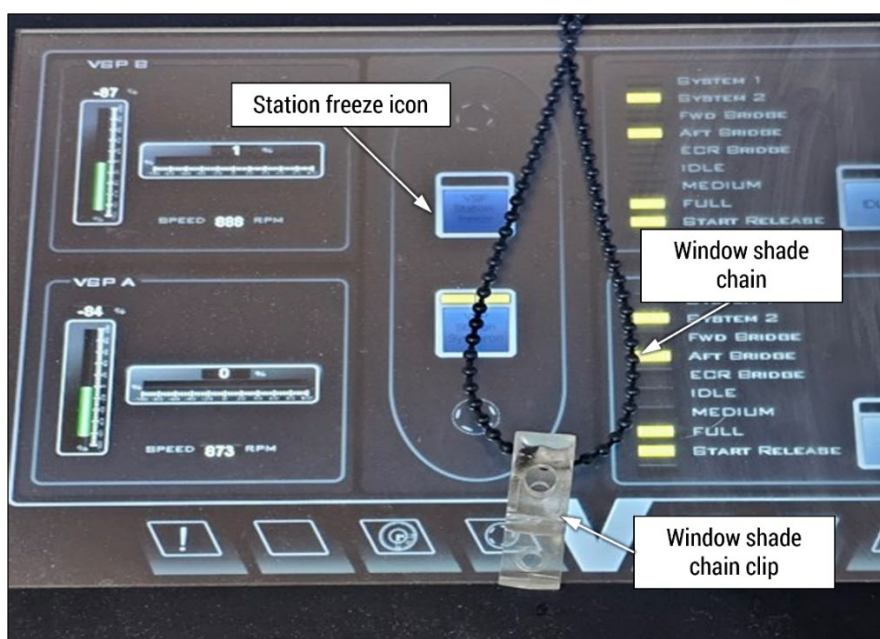
### **1.3.2 Postcasualty Troubleshooting**

On September 13, service technicians from the cycloidal propulsion system manufacturer reported to the *Ray Stoker, Jr.* to troubleshoot the loss of propulsion control that occurred on September 10. Technicians created an offline device for the system, checked the error file, and verified that no alarms were present during the time of control loss. According to the technician's report, it was unlikely that control was lost from both ends at the same time by a condition that did not cause alarms, and "only a complete loss of power or an extreme multi-point failure would cause such a loss without alarms if the system was faulty." Technicians and crew conducted sea trials for 3 days to attempt to replicate the loss of control. During the trials, the system operated as intended and no errors were observed.

The technicians concluded that the "station freeze" icon on one of the touch panels had likely been activated, which would have caused joystick control to be unresponsive without generating any alarms. Touching the station freeze icon would

have locked the joysticks in their last position and prevented any changes to the propulsion units if the joysticks were adjusted. The captain said he had never used the station freeze icon but understood it could be used to hold the vessel at a dock with a constant propulsion setting.

Technicians discovered that when the plastic clip on a window shade chain touched the station freeze icon on the touch panel, the station freeze function activated (see figure 6). The captain could not confidently recall the position of the window shade chains, nor whether he had adjusted any of the window shades during the transit (which would have required moving the chain for each shade). According to one deckhand, crewmembers would typically lower the shades when the sun was “too bright” and raise the shades if the sun was going down. The captain stated that, at times during previous transits, he had seen a window shade chain lying across the control touch panel, and he usually removed it from the controls.



**Figure 6.** Touch panel with plastic chain during postcasualty testing. (Source: Voith Service Report provided by Texas Department of Transportation)

Technicians instructed crewmembers that, when cleaning the touch panels, they should select a screen that would not have the ability to make operational changes. The captain stated that he did not clean any touch panels on the evening of the casualty.

### 1.3.3 Postcasualty Actions

As a result of the casualty and the postcasualty troubleshooting, *Ray Stoker, Jr.* crewmembers mounted clips on the bulkheads to hold the window shade chains in

place and prevent them from contacting the touch panels. Additionally, a sign was posted in each wheelhouse stating, "Please ensure blind lanyard does not come in contact with [touch panel] screen." The vessel's management team instructed crewmembers on three other ferries with similar touch panels to install fittings to secure the plastic clips for the window shades so that they would not come into contact with the station freeze icon on the control touch panels.

Management also updated their Vessel Safety Equipment Checklist to include checking the chains were secured in the clips, and sent a memorandum on September 23, 2024, to all captains and helmsmen, warning them of that the wheelhouse station freeze icon could be inadvertently activated and instructing them to ensure no items were placed over the touch panels. The memorandum also directed crewmembers to check the station freeze function icon in the event control was lost.

In July 2025, transparent plexiglass covers were installed on the *Robert H. Dedman*, which had similar control arrangement. In August 2025, transparent plexiglass covers were installed on the *Ray Stoker, Jr* (see figure 7). As of the date of this report, installation was still pending on the remaining two ferries with similar control arrangements.



**Figure 7.** Touch panel with plexiglass cover. (Background source: Texas Department of Transportation)

## 2 Analysis

While the vehicle and passenger ferry *Ray Stoker, Jr.* was underway from Bolivar to Galveston, the crew experienced a loss of cycloidal propulsion system control (thrust and steering control), and the ferry struck a floating pier at Coast Guard Base Galveston before grounding and coming to a stop.

During the loss of cycloidal propulsion system control, the captain adjusted both joysticks, but neither produced any speed or directional changes to either propulsion unit. Postcasualty inspection of the system by the manufacturer's service technicians indicated that no alarms activated during the incident. Three days of sea trials were conducted in an attempt to replicate the loss of control, but the system operated as intended. Therefore, the vessel's loss of cycloidal propulsion system control was not caused by an issue with the propulsion control system.

The control touch panel for the cycloidal propulsion system had a "station freeze" function that, when activated, would lock (or freeze) the joysticks in the last set position. Technicians considered the possibility that an object could have inadvertently activated this function. They tested the plastic clip on a window shade chain by touching the clip to the touch panel, and the clip activated functions in the same manner as an operator's finger. Crewmembers in the wheelhouse did not recall clicking the station freeze icon, and they stated that they had not cleaned the touch panels during the transit. The captain stated that, at times during previous transits, he had seen the wheelhouse window shade chain lying across a control touch panel, and he usually removed it from the panel. Based on the time of the casualty (1823—about an hour from sunset) and the number of crossings between Bolivar and Galveston, the shade chain could have been adjusted at some point due to the ferry's position relative to the sun. According to one deckhand, they typically adjusted the shade if the sun was "too bright" or going down. Adjusting the shade increased the potential that the chain and its plastic clip, once released after adjustment by the operator, would lay atop the touch panel. However, crewmembers in the wheelhouse did not recall adjusting the window shade during the transit. If the shade chain and its plastic clip had been left lying across the touch panel, the clip could have activated the station freeze function during the transit, unbeknownst to the operator. Following the accident, the operating company installed transparent plexiglass covers over the control touch panels on the *Ray Stoker, Jr.* and instructed ferry crewmembers to install fittings to secure window shade chains so that they would not come into contact with the panels.

## 3 Conclusions

### 3.1 Probable Cause

The National Transportation Safety Board determines that the probable cause of the contact of the ferry *Ray Stoker, Jr.* with a floating dock was likely a wheelhouse window shade chain clip activating the station freeze function on a propulsion control touch panel, resulting in a temporary loss of propulsion control.

## Vessel Particulars

| Vessel                     | <i>Ray Stoker, Jr.</i>                                        |
|----------------------------|---------------------------------------------------------------|
| Type                       | Passenger (Ferry)                                             |
| Owner/Operator             | Texas Department of Transportation (Government)               |
| Flag                       | United States                                                 |
| Port of registry           | Galveston, Texas                                              |
| Year built                 | 1996                                                          |
| Official number            | 1046230 (US)                                                  |
| IMO number                 | N/A                                                           |
| Classification society     | N/A                                                           |
| Length (overall)           | 263.3 ft (80.3 m)                                             |
| Breadth (max.)             | 65.3 ft (19.9 m)                                              |
| Draft (casualty)           | 11.0 ft (3.4 m)                                               |
| Tonnage                    | 1,082 GRT                                                     |
| Engine power; manufacturer | 2 × 1,500 hp (1,119 kW); EMD 12V 645 Series E6 diesel engines |

NTSB investigators worked closely with our counterparts from **Marine Safety Unit Texas City** throughout this investigation.

The National Transportation Safety Board (NTSB) is an independent federal agency charged by Congress with investigating every civil aviation accident in the United States and significant events in other modes of transportation—railroad, transit, highway, marine, pipeline, and commercial space. We determine the probable cause of the accidents and events we investigate, and issue safety recommendations aimed at preventing future occurrences. In addition, we conduct transportation safety research studies and offer information and other assistance to family members and survivors for any accident or event investigated by the agency. We also serve as the appellate authority for enforcement actions involving aviation and mariner certificates issued by the Federal Aviation Administration (FAA) and US Coast Guard, and we adjudicate appeals of civil penalty actions taken by the FAA.

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For more detailed background information on this report, visit the [NTSB Case Analysis and Reporting Online \(CAROL\) website](#) and search for NTSB accident ID DCA24FM062. Recent publications are available in their entirety on the [NTSB website](#). Other information about available publications also may be obtained from the website or by contacting—

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